KHULNA UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Courtesy:
Dr. Md. Selim Hossain

Department of Electrical and Electronic Engineering

Course Code : EE 4121

VLSI

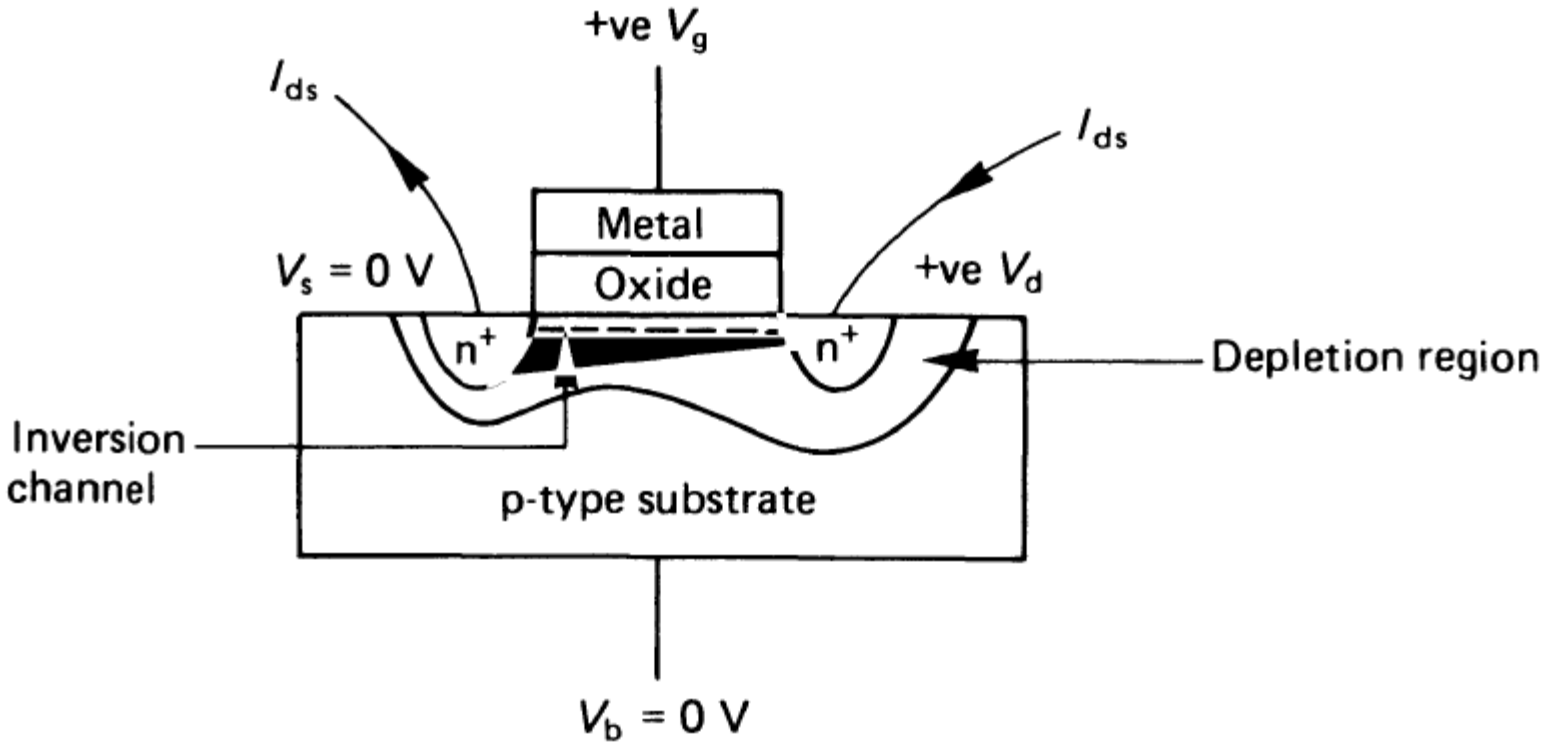


Figure 2.3 Conduction in an NMOS transistor

2.3 Threshold Voltage

The input potential V_{gs} at which the surface just becomes inverted is called the threshold voltage V_t. Below the threshold voltage an NMOS transistor is off and no current flows, while above the threshold the inversion channel is established and the device conducts.

A detailed analysis of the magnitude of the threshold voltage is complex and beyond the scope of this book. However, the factors determining V_t can be appreciated from a consideration of the excess charge within an NMOS device when the inversion layer is established. This is depicted in figure 2.4.

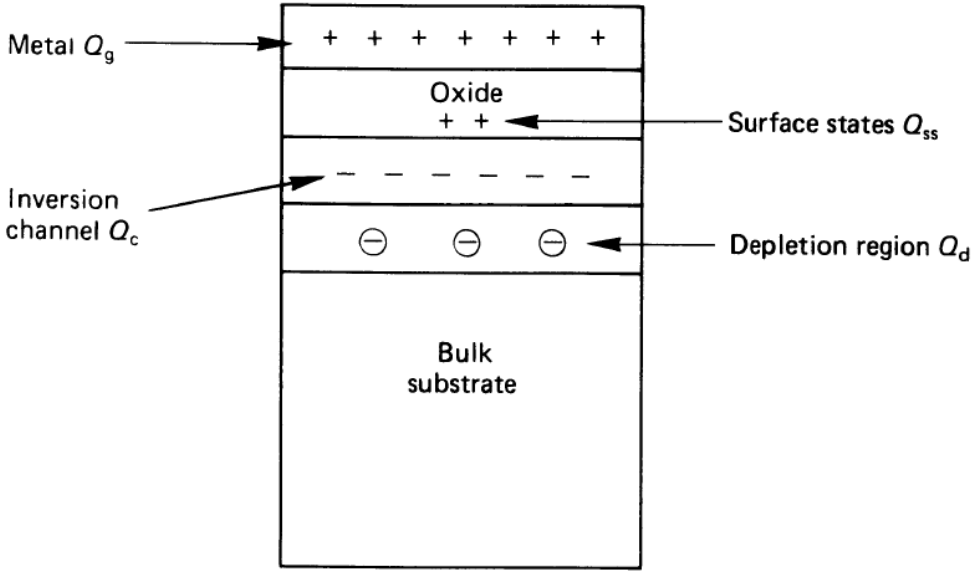


Figure 2.4 Excess charge within conducting NMOS transistor

$$Q_{ss} + Q_g = Q_c + Q_d$$

When the surface is just at the point of inversion, $Q_c = 0$ and

$$V_t = \frac{Q_g}{C_g} = \frac{Q_d - Q_{ss}}{C_g}$$

where C_g is the capacitance across the insulator.

Threshold Voltage

In practice, V_t has to overcome some in-built potential differences before the transistor is brought to the edge of conduction. As a result, another two voltage terms have to be included in the above expression for V_t. V_{dif} represents the voltage arising as a result of the difference between the gate and the semiconductor material. Since silicon gates are used nowadays, V_{dif} is small. The other voltage term, V_r, is the voltage across the depletion region just at the point of inversion; it is dependent upon the impurity concentration and is usually less than 1 volt. Thus

$$V_t = \frac{Q_d}{C_g} + V_r - \frac{Q_{ss}}{C_g} + V_{dif} \qquad (2.1)$$

The last three terms can be regarded as constant potentials, while Q_d is dependent upon transistor parameters and the applied voltages. In particular, Q_d is dependent upon the impurity concentration of the semiconductor material beneath the oxide. This provides a mechanism for adjusting the threshold voltage.

Threshold Voltage

For an NMOS device, the first two terms of equation (2.1) are positive and the last two negative, allowing the threshold to be made either positive or negative by suitable doping. The threshold is positive (usually 1 V) if the semiconductor surface between the source and the drain is heavily doped with a p-type impurity; such a device is off when the gate–source voltage is 0 V and is referred to as an 'enhancement mode NMOS transistor'.

The threshold can be made negative (typically −4 V) by doping the semiconductor surface with an n-type impurity. This has the effect of making the surface n-type, although the bulk semiconductor remains p-type. Thus even with $V_{gs} = 0$ V, the channel region is inverted and the device is on; an NMOS device which is on when V_{gs} is 0 V is called an 'NMOS depletion mode transistor'. Here, it is necessary to apply a negative gate–source voltage in order to repel electrons from the surface and turn the device off.

Threshold Voltage & body effect

Thresholds are quoted for transistors assuming a source–substrate voltage of 0 V. Changing the substrate voltage causes the threshold to change, and this effect is known as the 'body effect'. In NMOS devices, taking the substrate voltage negative of the source causes the depletion region surrounding the conduction path to widen and thus increases Q_d. As a result, V_t has to be increased to bring the transistor to the edge of conduction. If V_{t0} is defined as the threshold when the source–substrate voltage V_{sb} is 0 V, then the modified threshold V_t to take into account the body effect is

$$V_t = V_{t0} + \gamma(V_{sb})^{1/2}$$

γ is a constant dependent upon transistor parameters and tends to lie between 0.3 and 0.7 for MOS transistors. It is usual to use a value of 0.5 in calculations.

Similar reasoning for a PMOS transistor shows that increasing the substrate voltage above the source potential causes the threshold to become more negative.

2.4 I_{ds} versus V_{ds} Characteristic for NMOS Devices

Figure 2.5 shows the symbols used in this book to denote MOS transistors. It is assumed that the substrate of NMOS devices is tied to the most negative voltage available and a PMOS substrate to the most positive available voltage. Thus this connection is usually omitted from circuit diagrams.

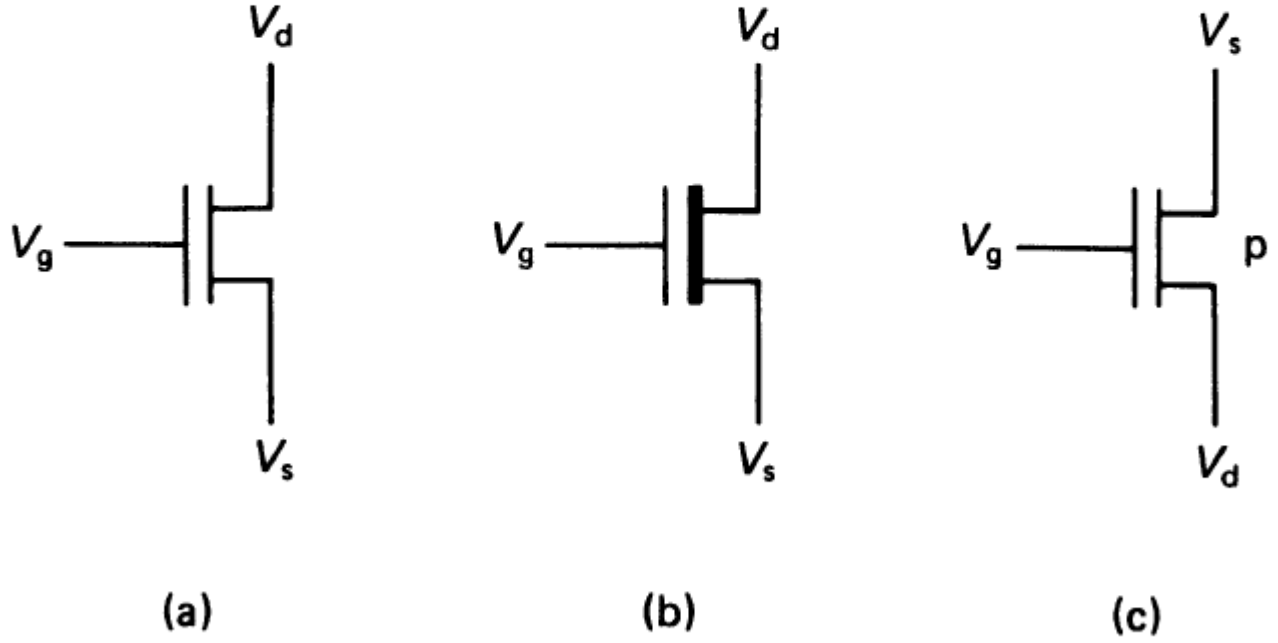


Figure 2.5 Symbols for MOS transistors: (a) enhancement mode NMOS, (b) depletion mode NMOS, (c) enhancement mode PMOS

Consider an NMOS device. When $V_{gs} < V_t$ the transistor is off, regardless of the drain voltage; the device does not conduct and no current flows. The device conducts when $V_{gs} > V_t$, and if a constant V_{gs} is applied then the resulting I_{ds} versus V_{ds} curve can be split into two regions.

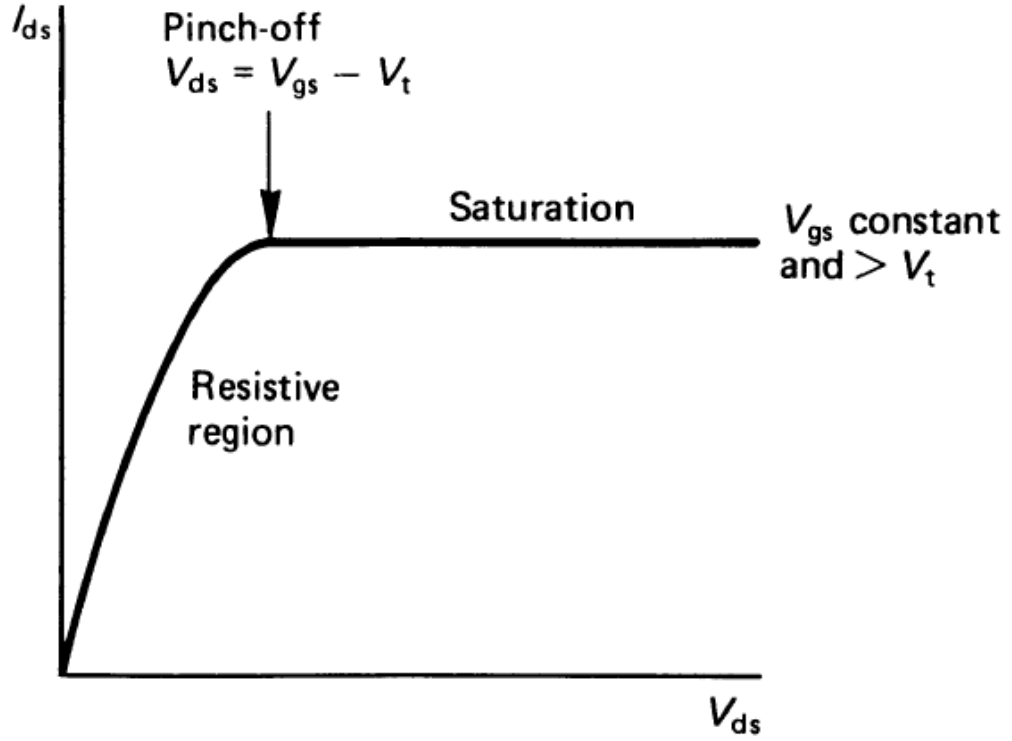


Figure 2.7 I_{ds} versus V_{ds} for an NMOS transistor

(a) Resistive Region

Here, $V_{ds} < V_{gs} - V_t$. The voltage across the insulator at the source is V_{gs} and at the drain is V_{gd} (meaning $V_g - V_d$). Although the voltage across the insulator is not constant, a voltage in excess of V_t exists at all points across the oxide, causing the formation of a continuous inversion channel between the drain and the source. It will be assumed that the increase in voltage along the channel from the drain to the source is linear with distance.

The device structure therefore resembles an infinite number of capacitances between the drain and source, each one having a different voltage across it and therefore a different charge from its neighbours. The total charge induced in the channel is the sum of the charge induced on each of these capacitances.

Consider one of these capacitances of length dx situated at a distance x metres from the drain, as shown in figure 2.6.

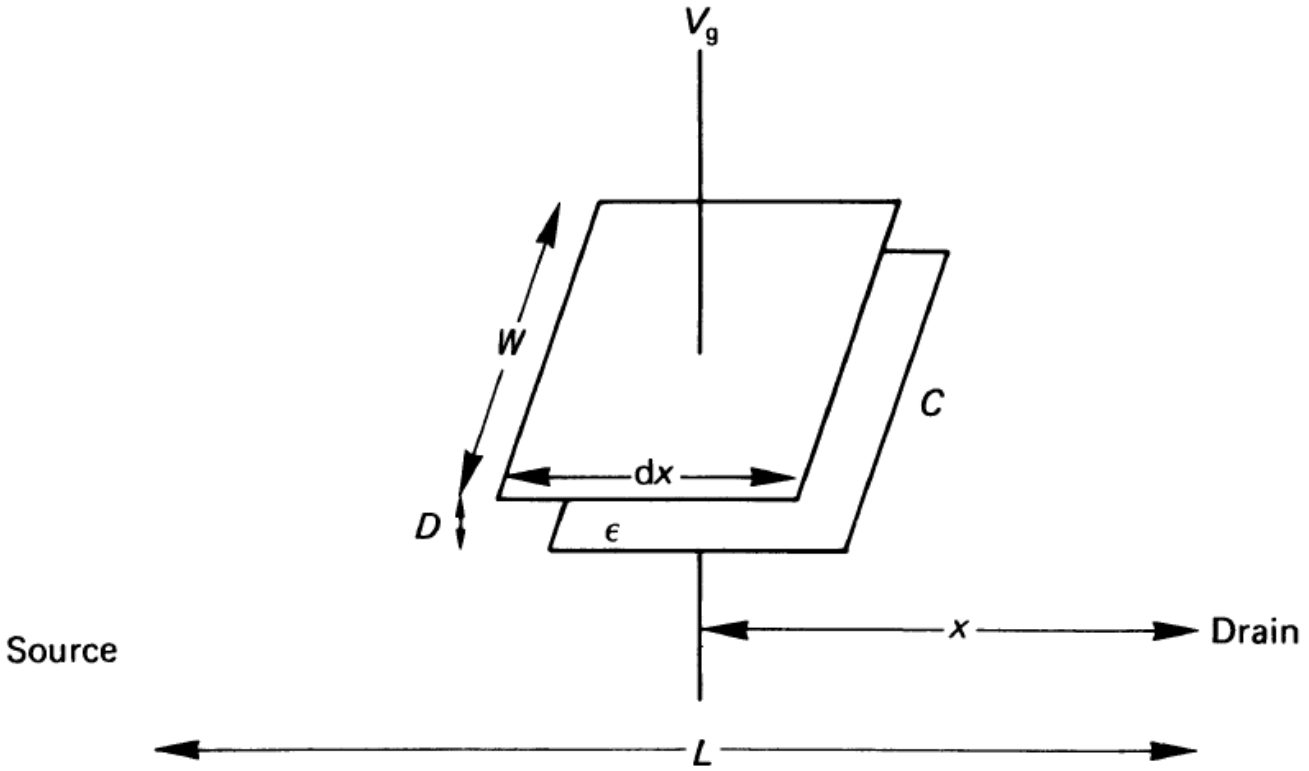


Figure 2.6 Elemental capacitor between source and drain

$$C = \frac{W\epsilon dx}{D}$$

where ϵ is the permittivity of the insulator in farads/metre and D is the thickness of the oxide in metres. The voltage v in excess of V_t across this capacitor is

$$v = (V_{gd} + \frac{x}{L}V_{ds} - V_t) = (V_{gs} - V_{ds} + \frac{x}{L}V_{ds} - V_t)$$

Thus the charge, q coulombs, induced on this capacitor is

$$q = C_v = \frac{W\epsilon dx}{D}(V_{gs} - V_{ds} + \frac{x}{L}V_{ds} - V_t)$$

The total charge Q induced in the channel is

$$Q = \int_0^L \frac{\epsilon W}{D}(V_{gs} - V_{ds} + \frac{x}{L}V_{ds} - V_t)dx$$

$$= \frac{\epsilon WL}{D}\left[(V_{gs} - V_t) - \frac{V_{ds}}{2}\right]$$

Now $Q = tI_{ds}$, where t is the time in seconds for an electron to move across the channel and

$$t = \frac{\text{channel length } L}{\text{electron velocity}}$$

μ_n is the electron velocity per unit electric field (measured in metre^2/volt-second) and is called the 'electron mobility'. Thus the electron velocity in the channel is $\mu_n V_{ds}/L$. Hence the current I_{ds} in amps is

$$I_{ds} = \frac{Q}{t} = \frac{\epsilon W \mu_n}{LD}\left[(V_{gs} - V_t)V_{ds} - \frac{V_{ds}^2}{2}\right] \tag{2.2}$$

For a constant V_{gs}, I_{ds} increases with an increase in V_{ds}, as shown in figure 2.7. It should be noted that at drain–source voltages which are very small compared with $V_{gs} - V_t$, the equation for the channel current reduces to

$$I_{ds} = \frac{\epsilon W \mu_n}{LD}(V_{gs} - V_t)V_{ds}$$

giving a linear relationship between I_{ds} and V_{ds} for a constant V_{gs}.

(b) Saturation

In this region $V_{ds} \geqslant V_{gs} - V_t$. As the drain voltage rises, the voltage across the insulator at the drain drops, and at $V_{ds} = V_{gs} - V_t$ it is V_t. This is the voltage necessary to just support inversion, and this point on the I_{ds} versus V_{ds} characteristic is called 'pinch-off'. At this point, the inversion channel ends just at the drain. As V_{ds} increases beyond pinch-off, the point at which inversion ceases moves away from the drain as shown in figure 2.8.

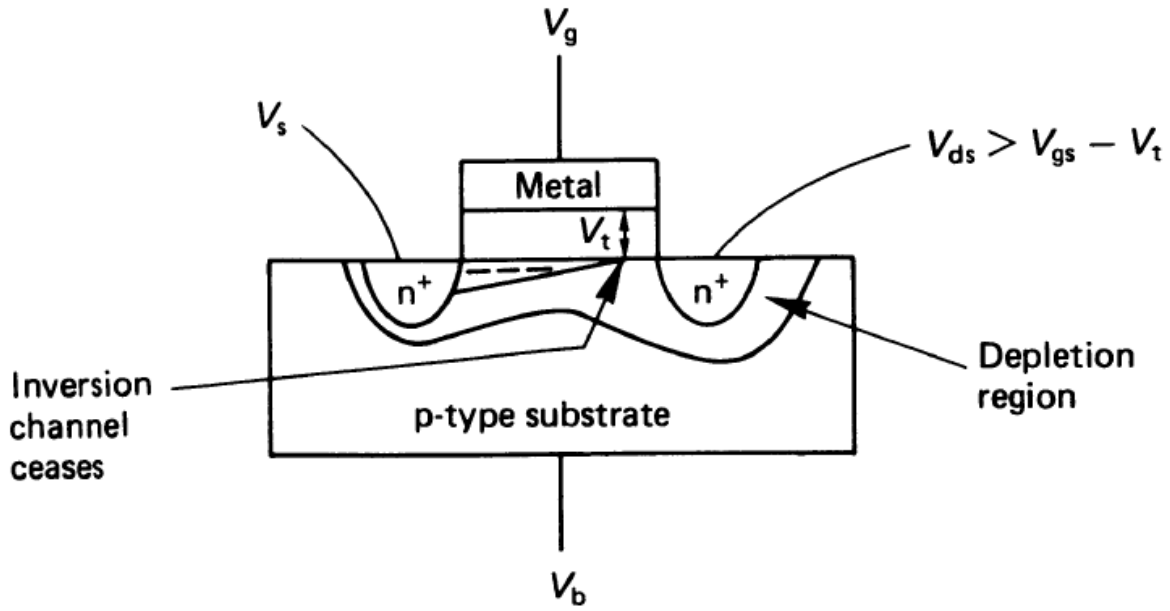


Figure 2.8 Inversion channel beyond pinch-off

The voltage difference along the inversion channel from the source to where it ceases is $V_{gs} - V_t$, and the excess potential $V_{ds} - V_{gs} + V_t$ is dropped between the end of the inversion channel and the drain. This creates a high electric field across this very short distance and the electrons from the inversion channel are quickly swept across this area to the drain.

At and above pinch-off the voltage between the source and the end of the inversion channel is constant at $V_{gs} - V_t$. The channel length can also be considered to be constant (equal to L). Thus the current flow is constant and the device is said to be 'saturated'. Replacing V_{ds} by $V_{gs} - V_t$ in equation (2.2), I_{ds} becomes

$$I_{ds} = \frac{\epsilon W \mu_n}{2LD}(V_{gs} - V_t)^2 \tag{2.3}$$

The I_{ds} versus V_{ds} characteristic for a constant V_{gs} is shown in figure 2.7. In practice, there is a slight increase in current with increasing V_{ds} above pinch-off, as a result of the reduction in the channel length.

2.5 Characteristic Equation for PMOS Devices

A PMOS enhancement device is off when V_{gs} is 0 V. Here it is necessary to take the gate voltage negative of the source in order to exceed the negative threshold and turn the transistor on. It is thus normal to connect the source terminal to a positive potential and operate the gate and the drain at voltages equal to or negative of this potential. Consequently current, I_{sd}, flows from the source to the drain in a PMOS device.

A similar analysis to that performed for NMOS devices allows the equations for I_{sd} versus V_{sd} to be obtained for the resistive and saturated regions. If the threshold voltage V_t for PMOS devices (only) is redefined to be the positive source-gate voltage at which the transistor just turns on, then equations similar to those for NMOS are obtained. In the resistive region where $V_{sd} < V_{sg} - V_t$

$$I_{sd} = \frac{\epsilon \mu_p W}{DL} \left[(V_{sg} - V_t) V_{sd} - \frac{V_{sd}^2}{2} \right] \tag{2.4}$$

Pinch-off occurs when $V_{sd} = V_{sg} - V_t$ and for $V_{sd} > V_{sg} - V_t$, the device is saturated. At and above pinch-off

$$I_{sd} = \frac{\epsilon \mu_p W}{2DL} (V_{sg} - V_t)^2 \qquad (2.5)$$

μ_p is the hole velocity per unit electric field and is known as the 'hole mobility'. It is two to three times less than μ_n. Since the current flow is proportional to the carrier mobility, an NMOS transistor will conduct more current than a PMOS device of similar size. This can be seen in figure 2.9 which shows characteristic curves for PMOS and NMOS transistors if typical values are assumed for the parameters.

The speed of a circuit is dependent upon the rate at which circuit capacitances can be charged and discharged. This in turn is dependent upon the current available from devices within the design. The current capability of PMOS devices can be made equal to that for NMOS devices by increasing their size to compensate for the difference between hole and electron mobility. However, these larger devices require a greater silicon area and have an increased circuit capacitance. For this reason, NMOS devices are used in preference to PMOS in circuit design.